🍔 3D Coordinates Extension — English Worksheet (Advanced)

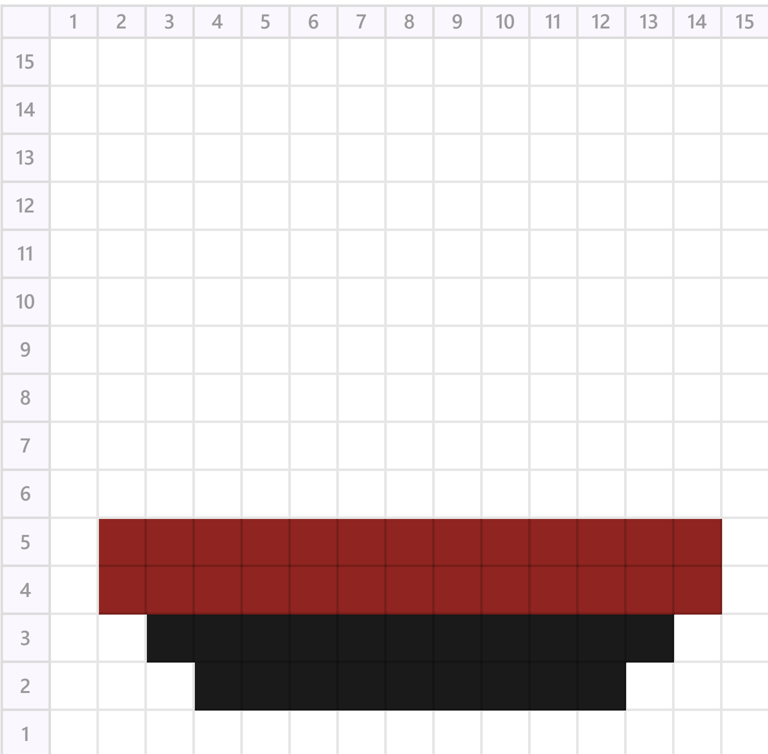
Topic: Finish the Burger (x, y) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Lesson 3) · **Time:** about 45 minutes

Open the world. One burger is **finished**. One is only **started** — it has a bottom bun and a tomato, nothing else.

Your job: **finish it**. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

> 🔴 Put a **red block at your feet** before you start. That is your **home spot**. Stand on it every run, or the burger moves too.

🍔 On the wall now — bottom bun · tomato



1 · Predict 🔮

Read each set of steps. Write what you will see **and the reason**.

```
place red block at (2, 4)
place red block at (3, 4)
place red block at (4, 4)
```

Three red blocks share y = 4. Do they make a line going across or up? Which number stays the same, and why?

```
place brown block at (8, 9)
place brown block at (8, 10)
place gold block at (8, 11)
place gold block at (8, 12)
```

These four share x = 8. Are they a row going across, or a column going up? Which number changes, and why?

```
place brown block at (5, 9)
place brown block at (6, 9)
place brown block at (5, 10)
place brown block at (6, 10)
```

These four blocks change both numbers. What shape do they make on the wall, and why?

```
place green block at (7, 7)
place yellow block at (7, 7)
```

Two blocks aim for the same (x, y). What happens on the wall — green, yellow, or both? Explain the reason.

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — The yellow cheese belongs **above** the lettuce, at **y = 8**. This block lands on the tomato at **y = 4**.

```
place yellow block at (2, 4)
```

Hint: y is **up**. Count the rows between the tomato and the cheese.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The bun's left edge should sit at **(2, 12)**. The two numbers are **swapped**, so it lands in the wrong place.

```
place gold block at (12, 2)
```

Hint: the **first** number is across (x), the **second** is up (y). Which one should be 2, which should be 12?

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📸 🎥

Stand on your **home spot** (the red block). Finish the burger block by block, reading each square's (x, y) — **x** across the top, **y** down the side. The cheese **drips** into the lettuce — copy the drips too.

🍔 **The finished burger, cheese melting — bun · gold, patty · brown, cheese · yellow, lettuce · green, tomato · red, crust · black**

Plan first. Find the **lowest** missing block and the **left-most** missing block. Write their (x, y) here. That tells you where your work starts.

Modify challenge. After the burger matches the picture, add **one topping of your own** — a new row, a sauce, a seed on the bun. Write the (x, y) of every block you add, then build it.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> The first thing I did was count from left to right. The first number was _____ and the last was _____. This is the **x** axis.
>
> The second thing I did was count down and up. The first number was _____ and the last was _____. This is the **y** axis.
>
> The hardest thing was building the _____.
>
> It was hard because _____.
>
> I think the burger looks _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.